Pearson Edexcel International A Level · Pure Mathematics 1

Chapter 3 — Equations and inequalities Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P1 第3章 · 方程与不等式
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1. P1-C03-R01 · P1 · Chapter 3

The line L has equation $y = 2x + 1$ and the curve C has equation $y = x^2 - 4x + 6$.

(a) Find the exact coordinates of the intersections of L and C. (3)

(b) Find the set of values of x for which the curve lies strictly below the line. (2)

(Total for Question 1 is 5 marks)

2. P1-C03-R02 · P1 · Chapter 3

An integer n satisfies $3(2n - 1) \leq 4n + 9$ and $5 - n < 8$.

(a) Find all possible values of n. (3)

(b) Find the greatest possible value of n^2 and state the value of n that gives it. (2)

(Total for Question 2 is 5 marks)

3. P1-C03-R03 · P1 · Chapter 3

The equations $y = x + 4$ and $xy = 12$ are to be solved simultaneously.

(a) Find both pairs (x, y). (3)

(b) Only solutions with $x > 0$ and $y > 0$ are permitted in a rectangular area model. State the permitted solution and find $2x + 2y$. (3)

(Total for Question 3 is 6 marks)

$x^2 - 7x + 10 \leq 0$.

(a) Solve the inequality. (2)

(b) Given also that $2x - 1 > 5$, find

(i) the resulting interval for x, (2)

(ii) all integers in this interval. (1)

(Total for Question 4 is 5 marks)

5.

The equations $x + y = 7$ and $x^2 + y^2 = 29$ hold.

(a) Find all ordered pairs (x, y). (3)

(b) Find the possible values of $x - y$. (2)

(Total for Question 5 is 5 marks)

6.

A delivery charge is £4 plus £1.50 per kilometre. A second service charges £10 plus £0.75 per kilometre. Let the distance be $d > 0$ kilometres.

(a) Find the distance at which the two charges are equal. (2)

(b) A customer has a budget of £19 and wants the second service to be strictly cheaper. Find the permitted interval for d. (3)

(Total for Question 6 is 5 marks)

7. P1-C03-R07 · P1 · Chapter 3

The line $y = 3x - 2$ meets the curve $y = x^2 + k$ at the point $(1, 1)$.

(a) Find k and the coordinates of the other intersection. (3)

(b) Find the values of x for which $x^2 + k \geq 3x - 2$. (2)

(Total for Question 7 is 5 marks)

8.

For a real number x, $4 - 3x \geq 2x - 11$.

(a) Solve the inequality. (2)

(b) Find all values of x that also satisfy $x^2 > 4$. (3)

(Total for Question 8 is 5 marks)

9. P1-C03-R09 · P1 · Chapter 3

The simultaneous equations are $2x + y = 5$ and $x^2 + xy = 6$.

(a) Find all real solutions (x, y). (3)

(b) A student divides the second equation by x before solving. Explain why this division does not remove a solution in this question. (2)

(Total for Question 9 is 5 marks)

The inequalities $x^2 + 2x - 8 > 0$ and $x < 3$ must both hold.

(a) Find the complete solution set. (3)

(b) State the greatest integer that satisfies both inequalities. (2)

(Total for Question 10 is 5 marks)

A rectangle has perimeter 24 m and area at least $32\ \text{m}^2$. Its width is w m and is no greater than its length.

(a) Show that $w^2 - 12w + 32 \leq 0$. (2)

(b) Find the permitted interval for w. (3)

(Total for Question 11 is 5 marks)

12. P1-C03-R12 · P1 · Chapter 3

The equations $y = 2 - x$ and $y = x^2 - 4$ have two intersections.

(a) Find the coordinates of both intersections. (3)

(b) For each intersection, calculate $x + y$ and explain why the answers agree. (3)

(Total for Question 12 is 6 marks)

$2x^2 + 3x - 9 < 0$.

(a) Solve the inequality. (2)

(b) Find the smallest and largest integers satisfying the inequality. (2)

(Total for Question 13 is 4 marks)

14.

A line $y = x + c$ passes through an intersection of the curves $y = x^2$ and $y = 4x - 3$.

(a) Find the two possible intersections of the curves. (2)

(b) Find the two possible values of c. (2)

(Total for Question 14 is 4 marks)

15.

The equations are $x - y = 1$ and $xy = 6$.

(a) Find all possible ordered pairs (x, y). (3)

(b) Find the unique solution satisfying $x + y < 0$ and state the value of $x^2 - y^2$ for this solution. (3)

(Total for Question 15 is 6 marks)

The expression $Q(x) = 9 - (x - 1)^2$.

(a) Find all values of x for which $Q(x) \geq 0$. (2)

(b) Given also that x is an integer and $2x + 1 < 6$, find

(i) all permitted values of x, (2)

(ii) the greatest permitted value of $Q(x)$. (2)

(Total for Question 16 is 6 marks)

17.

A ticket office sells adult tickets at £8 and child tickets at £5. A group buys 14 tickets for a total of £94.

(a) Form and solve simultaneous equations to find the numbers of adult and child tickets. (3)

(b) The same group size is used for a second booking with a budget of at most £85. Find the greatest possible number of adult tickets. (3)

(Total for Question 17 is 6 marks)

18.

The equations $y = x^2 - 2x$ and $y = 4 - x$ are to be solved simultaneously.

(a) Find the exact x-coordinates of their intersections. (3)

(b) Hence solve $x^2 - 2x < 4 - x$. (2)

(Total for Question 18 is 5 marks)

19.

A real number x satisfies $5 - 2(x + 1) < 3x + 8$.

(a) Solve this inequality. (2)

(b) Find the values of x that also satisfy $(x + 1)(x - 4) \leq 0$. (2)

(Total for Question 19 is 4 marks)

The line $x + y = 6$ and the curve $x^2 + 2y = 20$ intersect at A and B.

(a) Find the exact coordinates of A and B. (3)

(b) Find the midpoint of AB. (2)

(Total for Question 20 is 5 marks)

21. P1-C03-R21 · P1 · Chapter 3

A rectangle has side lengths $(x - 1)$ cm and $(9 - x)$ cm, both positive.

(a) Find the interval in which x must lie. (2)

(b) The area must exceed $12\ \text{cm}^2$. Find the permitted interval for x, giving exact endpoints. (3)

(Total for Question 21 is 5 marks)

For real x, $x(x-5) \geq 6$.

(a) Solve the inequality. (2)

(b) A positive integer x must also satisfy $3x + 2 < 26$. Find all its possible values. (2)

(Total for Question 22 is 4 marks)

23.

The equations are $y = 3x$ and $x^2 + y^2 = 40$.

(a) Find all real solutions. (3)

(b) Find the distance between the two solution points. (3)

(Total for Question 23 is 6 marks)

A function is defined by $f(x) = x^2 - 3x - 10$.

(a) Solve $f(x) \leq 0$. (2)

(b) Find all real x for which both $f(x) \leq 0$ and $f(-x) \leq 0$. (3)

(Total for Question 24 is 5 marks)

The equations $x + 2y = 1$ and $x^2 - 4y^2 = 9$ hold.

(a) Without forming a quadratic equation, find $x - 2y$. (2)

(b) Hence find x and y. (3)

(Total for Question 25 is 5 marks)

For a school event, the profit P pounds is modelled by $P = -n^2 + 18n - 65$, where n is a non-negative integer number of tickets sold.

(a) Find all values of n for which the profit is positive. (3)

(b) Find the least number of tickets that gives a profit of at least £15. (3)

(Total for Question 26 is 6 marks)

27. P1-C03-R27 · P1 · Chapter 3

The inequalities $2x^2 - 5x - 3 \geq 0$ and $x^2 < 16$ must hold simultaneously.

(a) Solve the first inequality. (2)

(b) Find the complete simultaneous solution set. (3)

(Total for Question 27 is 5 marks)

The equations $x + y = k$ and $x^2 + y^2 = 10$ have a solution with $x = y$.

(a) Find all possible values of k. (3)

(b) For $k = 2$, find all real solutions (x, y). (3)

(Total for Question 28 is 6 marks)

A real number x satisfies $2 \leq 3 - x < 7$.

(a) Express the solution as a single interval. (2)

(b) Find all values of x in this interval for which $x^2 + x - 6 > 0$. (3)

(Total for Question 29 is 5 marks)

The line $L : y = 2x + k$ meets the curve $C : y = x^2 - 2x + 3$ at two points with positive x-coordinates in the ratio $1 : 3$.

(a) Find k and the coordinates of the intersections. (4)

(b) For this value of k, consider points with $x > 2$.

(i) Find the values of x for which C lies on or above L. (3)

(ii) Find the least integer x in this range and the vertical separation of the curves at this input. (3)

(Total for Question 30 is 10 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P1-C03-R01 · 5 marks

(a) · 3 marks

$(1, 3)$ and $(5, 11)$.

联立得$x^2-6x+5=0$，分解为$(x-1)(x-5)$；代入直线求y。

M1	消去y
M1	解两根
A1	两交点

(b) · 2 marks

$1 < x < 5$.

差为$(x-1)(x-5)$，开口向上，在两根之间负；严格不等号排除端点。

M1	建立曲线与直线差的不等式
A1	严格区间

Question 2

P1-C03-R02 · 5 marks

(a) · 3 marks

$n = -2, -1, 0, 1, 2, 3, 4, 5, 6.$

第一式给$n \leqslant 6$，第二式给$n > -3$，交集中取整数。

M1	解第一不等式
M1	解第二不等式
A1	全部整数

(b) · 2 marks

36, at $n = 6$.

可行整数中绝对值最大的是6；平方36。

M1	在可行整数中比较绝对值
A1	最大值与取值

Question 3

P1-C03-R03 · 6 marks

(a) · 3 marks

$(2, 6)$ and $(-6, -2)$.

代入得$x^2+4x-12=0$，即$(x+6)(x-2)=0$；回代求y。

M1	代入消元
M1	分解求两根
A1	两对解

(b) · 3 marks

$(2, 6)$; $2x + 2y = 16$.

排除两负的解；把正解代入周长式。

B1	正确选择
M1	代入所给式
A1	结果16

Question 4

P1-C03-R04 · 5 marks

(a) · 2 marks

$2 \le x \le 5$.

分解为(x−2)(x−5)，向上抛物线在两根间及根上非正。

M1	找两个根
A1	闭区间

(b) Given also that $2x - 1 > 5$, find

(i) · 2 marks

$3 < x \le 5$.

新条件给x>3，与(a)交集。

M1	解新不等式
A1	正确交集

(ii) · 1 marks

$4, 5$.

3被排除，5被包含。

B1	两个整数

Question 5

P1-C03-R05 · 5 marks

(a) · 3 marks

$(2,5)$ and $(5,2)$.

置y=7−x得$2x^2-14x+20=0$，约简后(x−2)(x−5)=0。

M1	正确消元
M1	化简并解方程
A1	两个有序对

(b) · 2 marks

$-3,3$.

两个有序解分别相减，不可只保留正差。

M1	用两组解
A1	两个差

Question 6

P1-C03-R06 · 5 marks

(a) · 2 marks

$d = 8$ km.

$4 + 1.5d = 10 + 0.75d$，移项得0.75d=6。

M1	建立等价方程
A1	距离8公里

(b) · 3 marks

$8 < d \leq 12$.

第二服务较便宜要求d>8；预算要求10+0.75d⩽19，即d⩽12；取交集。

M1	严格比较费用
M1	预算不等式
A1	完整区间

Question 7

P1-C03-R07 · 5 marks

(a) · 3 marks

$k = 0$; other intersection $(2, 4)$.

代入已知点得1=1+k；再解$x^2=3x-2$，得x=1,2。

M1	用点求k
M1	联立求另一根
A1	k和另一坐标

(b) · 2 marks

$x \leq 1$ or $x \geq 2$.

代入k后$(x-1)(x-2)\geqslant 0$；取外侧并包括端点。

M1	正确分解
A1	两个闭外区间

Question 8

P1-C03-R08 · 5 marks

(a) · 2 marks

$x \leq 3$.

移项得15≥5x，除以正数5。

M1	正确移项
A1	解集

(b) · 3 marks

$x < -2$ or $2 < x \leq 3$.

平方大于4给x<−2或x>2；与x≤3求交，平方的两个端点均不包含。

M1	正确两支平方不等式
M1	求交集
A1	完整端点

Question 9

P1-C03-R09 · 5 marks

(a) · 3 marks

$(2, 1)$ and $(3, -1)$.

y=5−2x，代入得−x²+5x=6，即x²−5x+6=0。

M1	正确代入
M1	解二次方程
A1	两解

(b) · 2 marks

$x = 0$ cannot satisfy $x^2 + xy = 6$, since its left side would be 0.

先核查可能被除去的x=0，发现不满足原式，因此此题除x合法。

M1	核查被除因子为零的情况
A1	与右边6矛盾并作结论

Question 10

P1-C03-R10 · 5 marks

(a) · 3 marks

$x < -4$ or $2 < x < 3$.

二次式分解(x+4)(x−2)，取外侧，再与x<3求交。

M1	正确根及符号
M1	求交
A1	两个严格区间

(b) · 2 marks

-5.

区间(2,3)没有整数；另一支x<−4的最大整数为−5。

M1	分别检查两个分支中的整数
A1	最大整数

Question 11

P1-C03-R11 · 5 marks

(a) · 2 marks

$w(12-w)\geq 32 \Rightarrow w^2-12w+32\leq 0.$

另一边为12−w；移项再乘−1需翻转不等号。

M1	正确面积不等式
A1	负号转换正确

(b) · 3 marks

$4\leq w\leq 6.$

二次不等式给4⩽w⩽8；宽不大于长给w⩽6；取交集。

M1	解二次不等式
M1	使用宽长关系
A1	最终区间

Question 12

P1-C03-R12 · 6 marks

(a) · 3 marks

$(-3, 5)$ and $(2, 0)$.

联立得$x^2+x-6=0=(x+3)(x-2)$。

M1	正确消元
M1	解两根
A1	两个坐标

(b) · 3 marks

$x + y = 2$ at both points; both lie on $y = 2 - x$.

直线方程等价于$x+y=2$，所以无论曲线上的哪一个交点都满足这个和。

B1	两个和均正确
M1	使用直线方程
A1	完整解释

Question 13

P1-C03-R13 · 4 marks

(a) · 2 marks

$-3 < x < \frac{3}{2}$.

分解为(2x−3)(x+3)，向上开口在两根之间负。

M1	正确因式分解或求根
A1	严格区间

(b) · 2 marks

Smallest -2; largest 1.

不包含−3，也不允许超过或等于1.5；可行整数−2、−1、0、1。

M1	正确处理两个端点
A1	最小最大整数

Question 14

P1-C03-R14 · 4 marks

(a) · 2 marks

$(1, 1)$ and $(3, 9)$.

$x^2-4x+3=0$，解x=1,3并计算y。

M1	联立并求根
A1	两点

(b) · 2 marks

$c = 0, 6$.

由c=y−x，分别用两个交点计算。

M1	使用c=y-x
A1	两值

Question 15

P1-C03-R15 · 6 marks

(a) · 3 marks

$(3, 2)$ and $(-2, -3)$.

x=y+1给y^2+y−6=0，分解后y=2或−3。

M1	正确代入
M1	两根及回代
A1	两对

(b) · 3 marks

$(-2, -3)$; $x^2 - y^2 = -5$.

两组和分别5和−5，故选负组；平方差为4−9。

M1	正确筛选
A1	解对
A1	平方差

Question 16

P1-C03-R16 · 6 marks

(a) · 2 marks

$-2 \leq x \leq 4$.

转成$(x-1)^2 \leqslant 9$，得$-3 \leqslant x-1 \leqslant 3$。

M1	正确平方范围
A1	闭区间

(b) Given also that x is an integer and $2x + 1 < 6$, find

(i) · 2 marks

$-2, -1, 0, 1, 2$.

新条件$x<2.5$，与(a)求交再取整数。

M1	联立并取整数
A1	全部值

(ii) · 2 marks

9, at $x = 1$.

配方最大9，且$x=1$在可行整数内。

M1	检查最大值位置可行
A1	最大值

Question 17

P1-C03-R17 · 6 marks

(a) · 3 marks

8 adult and 6 child tickets.

令成人a、儿童c，a+c=14，8a+5c=94；相减得3a=24。

M1	两个方程
M1	正确消元
A1	两个票数

(b) · 3 marks

5 adult tickets.

费用8a+5(14−a)=70+3a≤85给a≤5，结合整数与0≤a≤14。

M1	正确费用不等式
M1	解并考虑整数
A1	最大人数

Question 18

P1-C03-R18 · 5 marks

(a) · 3 marks

$x = \dfrac{1 \pm \sqrt{17}}{2}$.

联立得到$x^2-x-4=0$，用公式解。

M1	正确方程
M1	二次公式
A1	两个精确根

(b) · 2 marks

$\dfrac{1 - \sqrt{17}}{2} < x < \dfrac{1 + \sqrt{17}}{2}$.

二次差开口向上，严格为负的位置在两根之间，不包含端点。

M1	使用根和开口确定符号
A1	精确区间

Question 19

P1-C03-R19 · 4 marks

(a) · 2 marks

$x > -1$.

展开得3−2x<3x+8，故−5<5x。

M1	正确展开移项
A1	解集

(b) · 2 marks

$-1 < x \leq 4$.

二次条件给−1⩽x⩽4；首条件在−1严格，所以交集左开右闭。

M1	正确二次区间
A1	正确端点交集

Question 20

P1-C03-R20 · 5 marks

(a) · 3 marks

$(-2, 8)$ and $(4, 2)$.

y=6−x给$x^2-2x-8=0$，因式分解求x=−2,4并回代。

M1	代入消元
M1	求根
A1	两点

(b) · 2 marks

$(1, 5)$.

横纵坐标分别求平均。

M1	对两坐标取平均
A1	中点

Question 21

P1-C03-R21 · 5 marks

(a) · 2 marks

$1 < x < 9$.

两边正分别要求x>1与x<9。

M1	建立两条正值条件
A1	交集

(b) · 3 marks

$3 < x < 7$.

面积16−(x−5)²>12给(x−5)²<4，故3

M1	建立正确面积式
M1	解平方不等式
A1	正确区间

Question 22

P1-C03-R22 · 4 marks

(a) · 2 marks

$x \leq -1$ or $x \geq 6$.

移项后(x−6)(x+1)⩾0，取外侧含根。

M1	正确分解
A1	两外区间

(b) · 2 marks

$x = 6, 7$.

新条件x<8；只保留(a)中的正分支x⩾6并取整数。

M1	联立正值与线性限制
A1	全部整数

Question 23

P1-C03-R23 · 6 marks

(a) · 3 marks

$(2, 6)$ and $(-2, -6)$.

代入得$10x^2=40$，所以$x=\pm 2$，y与x同号。

M1	正确消元
M1	取正负平方根
A1	两对解

(b) · 3 marks

$4\sqrt{10}$.

横差4、纵差12，距离$\sqrt{(16+144)}=4\sqrt{10}$。

M1	使用两点距离
M1	化简根式
A1	精确距离

Question 24

P1-C03-R24 · 5 marks

(a) · 2 marks

$-2 \leq x \leq 5$.

分解为(x−5)(x+2)，开口向上。

M1	正确根与符号
A1	闭区间

(b) · 3 marks

$-2 \leq x \leq 2$.

将−x代入(a)的条件得−5≤x≤2；和原区间求交。

M1	正确处理替换后的不等号
M1	求交集
A1	最终区间

Question 25

P1-C03-R25 · 5 marks

(a) · 2 marks

$x - 2y = 9$.

平方差为(x−2y)(x+2y)=9，用给定第二因子1得到结论。

M1	正确平方差因式
A1	求得线性组合

(b) · 3 marks

$x = 5,\ y = -2$.

把x+2y=1与x−2y=9相加得2x=10，再回代。

M1	联立两线性组合
A1	x
A1	y

Question 26

P1-C03-R26 · 6 marks

(a) · 3 marks

$n = 6, 7, 8, 9, 10, 11, 12.$

正利润给$n^2-18n+65<0$，即$(n-5)(n-13)<0$；取5

M1	正确转换不等式
M1	求根与区间
A1	全部整数

(b) · 3 marks

8 tickets.

$P=16-(n-9)^2$，要求$(n-9)^2\leqslant 1$，n=8、9、10，最少8。

M1	正确利润门槛关系
M1	取可行整数
A1	最小值

Question 27

P1-C03-R27 · 5 marks

(a) · 2 marks

$x \le -\frac{1}{2}$ or $x \ge 3$.

因式为(2x+1)(x−3)，乘积非负取外侧。

M1	正确因式分解
A1	两外区间

(b) · 3 marks

$-4 < x \le -\frac{1}{2}$ or $3 \le x < 4$.

第二条件为−4

M1	正确平方范围
M1	对两个分支求交
A1	全部端点

Question 28

P1-C03-R28 · 6 marks

(a) · 3 marks

$k = \pm 2\sqrt{5}$.

代入x=y得$2x^2=10$，所以x=y=±√5且同号；k=2x。

M1	使用相等条件
M1	保留两个符号
A1	两k值

(b) · 3 marks

$(-1, 3)$ and $(3, -1)$.

y=2−x给$2x^2-4x-6=0$，解x=−1或3。

M1	正确消元
M1	解方程
A1	两对

Question 29

P1-C03-R29 · 5 marks

(a) · 2 marks

$-4 < x \leq 1.$

分开两侧：x≤1与x>−4，乘除负数时不等号翻转。

M1	正确处理两侧负号
A1	正确区间

(b) · 3 marks

$-4 < x < -3.$

二次式(x+3)(x−2)>0给x<−3或x>2；后者与(a)无交。

M1	解二次不等式
M1	求交
A1	严格最终区间

Question 30

P1-C03-R30 · 10 marks

(a) · 4 marks

$k = 0$; $(1, 2), (3, 6)$.

令两根t、3t，联立式$x^2-4x+3-k=(x-t)(x-3t)$。比较一次项4t=4得t=1，再比较常数$3-k=3$，最后代直线求y。

M1	线曲联立
M1	用根比比较系数
M1	求k并回代
A1	参数和两点

(b) For this value of k, consider points with $x > 2$.

(i) · 3 marks

$x \geq 3$.

差为$(x-1)(x-3)$，非负取$x\leqslant 1$或$x\geqslant 3$，再与$x>2$求交。

M1	正确二次不等式
M1	与新条件求交
A1	范围

(ii) · 3 marks

$x = 3$; separation 0.

闭端点3可取，恰好是交点所以纵向分离为0。

B1	最小整数
M1	在该输入计算差
A1	分离量